

properties of sine and cosine

(c) even and odd properties

$$\cos(-x) = \cos x, \sin(-x) = -\sin x$$

proof:

$$\textcircled{1} \because \cos(y-x) = \cos y \cos x + \sin y \sin x$$

let $y=0$

$$\therefore \cos(-x) = \cos 0 \cos x + \sin 0 \sin x$$

$$\therefore \cos(-x) = \cos x$$

$$\textcircled{2} \because \cos(y-x) = \cos y \cos x + \sin y \sin x$$

$$= -\sin x$$

$$\therefore \cos(\frac{\pi}{2}-x) = \cos \frac{\pi}{2} \cos x + \sin \frac{\pi}{2} \sin x = \sin x$$

$$\textcircled{3} \sin(-x) = \cos(\frac{\pi}{2}+x)$$

$$= \sin x$$

$$\textcircled{2} \because \cos(\frac{\pi}{2}-x) = -\sin x$$

$$\therefore \cos(\pi+x) = \cos(\frac{\pi}{2}+\frac{\pi}{2}+x) = -\sin(\frac{\pi}{2}+x) = -\cos x$$

$$\therefore \cos(2\pi+x) = \cos(\pi+\pi+x) = -\cos(\pi+x) = \cos x$$

(d) co-relations:

$$\sin(\frac{\pi}{2}+x) = \cos x$$

$$\cos(\frac{\pi}{2}+x) = -\sin x$$

$$\text{proof: } \textcircled{1} \because \cos(\frac{\pi}{2}-x) = \sin x$$

$$\therefore \cos(\frac{\pi}{2}-(\frac{\pi}{2}+x)) = \sin(\frac{\pi}{2}+x)$$

$$\therefore \sin(\frac{\pi}{2}+x) = \cos(-x) = \cos x$$

(f) addition formulas

$$\cos(x+y) = \cos x \cos y - \sin x \sin y$$

$$\sin(x+y) = \sin x \cos y + \cos x \sin y$$

$$\text{proof: } \textcircled{1} \because \cos(y-x) = \cos y \cos x + \sin y \sin x$$

$$\therefore \cos(x+y) = \cos y \cos(-x) + \sin y \sin(-x)$$

$$= \cos y \cos x - \sin y \sin x$$

1. axioms

(1) $\sin, \cos$ defined on $\mathbb{R}$

domain of definition:

(2) special values: $\cos 0 = 1, \sin \frac{\pi}{2} = 1, \cos \pi = -1$

(3) cosine of a difference: $\cos(y-x) = \cos y \cos x + \sin y \sin x, \forall x, y$

(4) fundamental inequalities

$$0 < \cos x < \frac{x}{\pi} < \frac{1}{\pi}, \forall x \in (0, \frac{\pi}{2})$$

2. theorem

(a) Pythagorean identity: $\sin^2 x + \cos^2 x = 1, \forall x$

proof: $\because \cos(y-x) = \cos y \cos x + \sin y \sin x, \forall x, y.$
let $y=x,$

$$\therefore \sin^2 x + \cos^2 x = 1$$

(b) special values: $\sin \frac{\pi}{2} = 1, \cos \frac{\pi}{2} = 0$

proof: $\because \sin x + \cos^2 x = 1,$

$$\therefore \cos^2 x = 1, \therefore \sin x = 0$$

$$\therefore \cos \pi = -1, \therefore \sin \pi = 0$$

$$\therefore \sin \frac{\pi}{2} = 1, \therefore \cos \frac{\pi}{2} = 0$$

(h) monotonicity

$\sin x$ on $[0, \frac{\pi}{2}]$, $x \downarrow$ on $[0, \frac{\pi}{2}]$

proof:

$$\because 0 < x < \frac{\pi}{2} < \frac{1}{c}x, \forall x \in (0, \frac{\pi}{2})$$

$$\because cx > 0, sx > 0, \forall x \in (0, \frac{\pi}{2})$$

if $0 < b < a < \frac{\pi}{2}$,

$$\because 0 < \frac{a+b}{2} < \frac{\pi}{2}$$

$$0 < \frac{a-b}{2} < \frac{\pi}{2} < \frac{\pi}{2}$$

$$\because \frac{a+b}{2} > 0, \frac{a+b}{2} > 0, \frac{a+b}{2} > 0$$

$$\because sa > sb, ca < cb, (by g)$$

$$\textcircled{5} \because sx = -c(\frac{\pi}{2} + x)$$

$$\therefore s(x+y) = -c(\frac{\pi}{2} + x+y)$$

$$\begin{aligned} &= -c(\frac{\pi}{2} + x)cy + s(\frac{\pi}{2} + x)sy \\ &= sxcy + cxsy \end{aligned}$$

(g) difference formulas

$$sa - sb = 2s\frac{a+b}{2}c\frac{a-b}{2}$$

$$ca - cb = -2s\frac{a+b}{2}s\frac{a-b}{2}$$

$$\text{proof: } \textcircled{1} \because s(x+y) = sxcy + cxsy$$

$$s(x-y) = sxcy - cxsy$$

$$\therefore s(x+y) - s(x-y) = 2cxsy$$

$$\textcircled{5} \because c(x+y) = cxcy - sxsy$$

$$c(x-y) = cxcy + sxsy$$

$$\therefore c(x+y) - c(x-y) = -2sxsy$$

$$\text{let } x = \frac{a+b}{2}, y = \frac{a-b}{2}$$

$$\therefore sa - sb = 2c\frac{a+b}{2}s\frac{a-b}{2}$$

$$ca - cb = -2s\frac{a+b}{2}s\frac{a-b}{2}$$